

Github Tasks

Task 1: Create a GitHub account.

Objective:

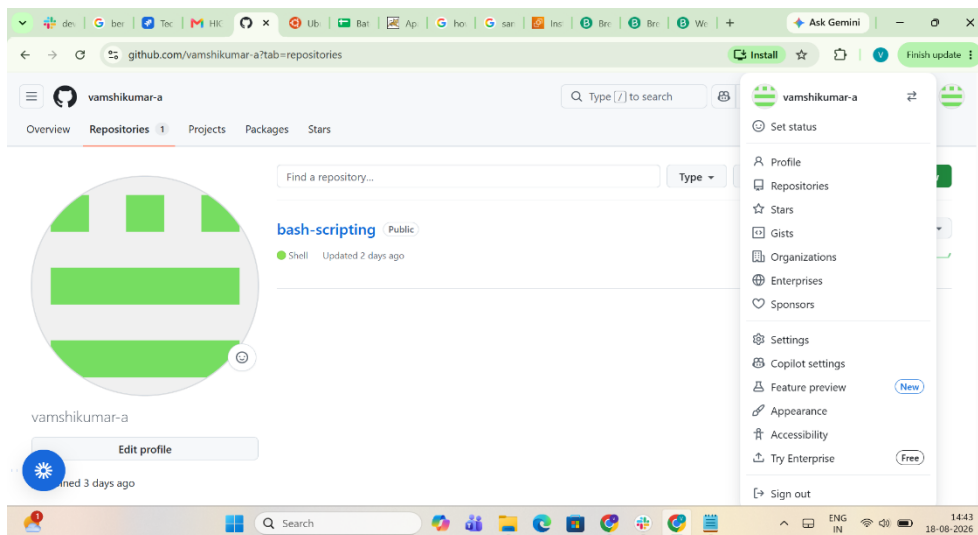
Create a GitHub account for hosting and managing source-code repositories.

Purpose:

To create a GitHub account that can be used for version control, code management, and collaboration.

Implementation:

1. Opened the GitHub website.
2. Clicked **Sign up** to create a new account.
3. Entered the email address and created a strong password.
4. Selected a unique GitHub username.
5. Completed the account verification process.
6. Verified the email address associated with the GitHub account.
7. Logged in and verified the newly created GitHub account.



Conclusion:

Successfully created and verified a GitHub account.

Task 2: Set up MFA (Multi-Factor Authentication) for the GitHub account using a mobile app.

Objective:

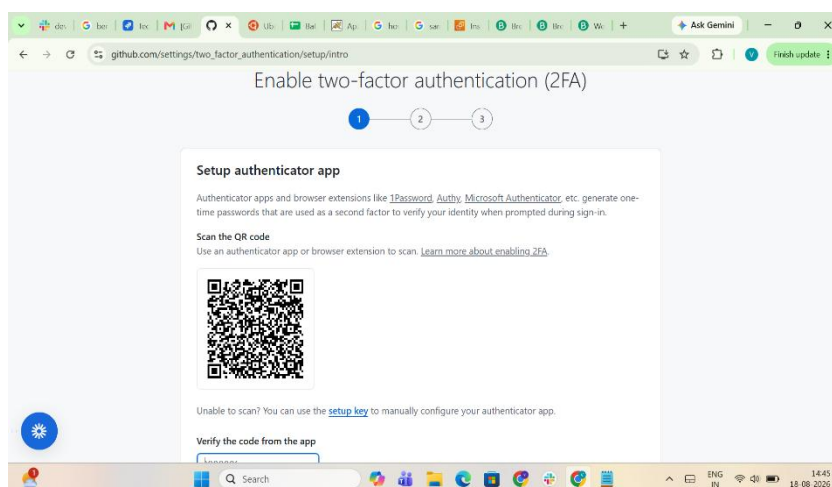
Enable Multi-Factor Authentication (MFA) for the GitHub account using a mobile authenticator application.

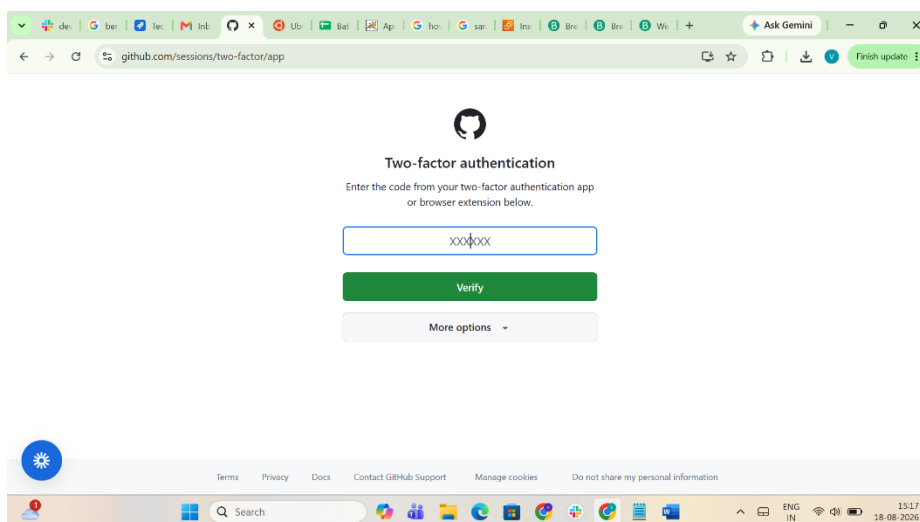
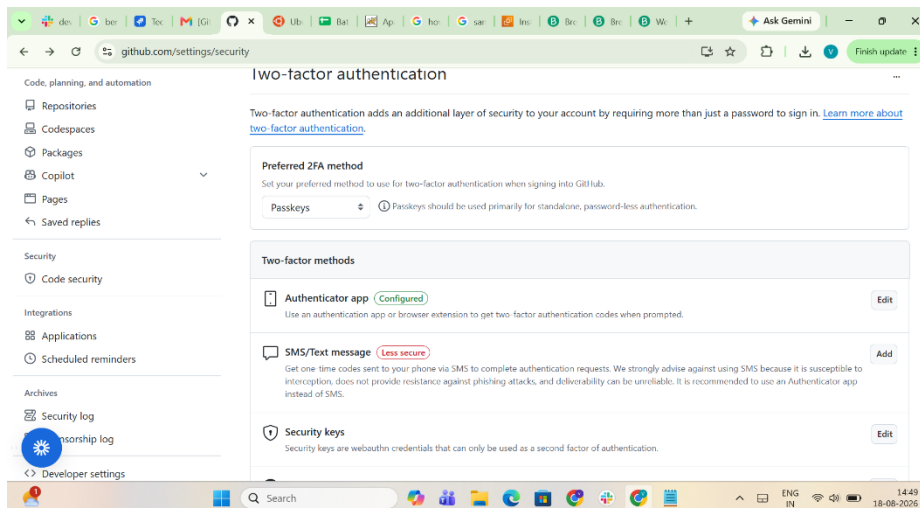
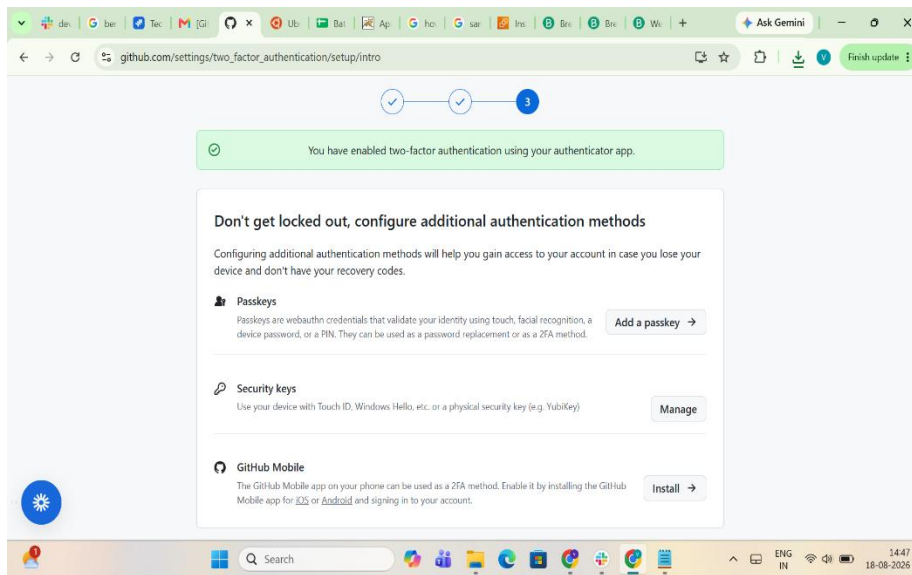
Purpose:

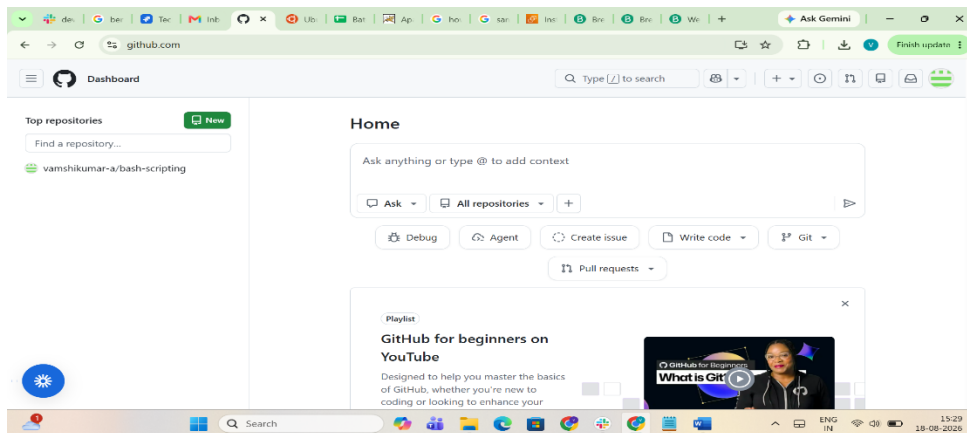
To improve GitHub account security by requiring an additional authentication code along with the password.

Implementation:

1. Logged in to the GitHub account.
2. Opened **Settings** from the GitHub profile menu.
3. Navigated to **Password and authentication**.
4. Selected the option to enable **Two-Factor Authentication (2FA)**.
5. Selected the **Authenticator App** method.
6. Installed an authenticator application on the mobile phone.
7. Scanned the QR code provided by GitHub using the authenticator app.
8. Entered the verification code generated by the mobile authenticator.
9. Completed the MFA setup and verification.
10. Saved the recovery codes securely.
11. Logged out and logged in again to verify MFA authentication.







Conclusion:

Successfully enabled and verified MFA using a mobile authenticator application, providing an additional layer of security to the GitHub account.

GitHub commands used in video

1. `git --version` → Checks Git version.

```
91738@IDEAPAD MINGW64 ~/Desktop/hiring-app
$ git --version
git version 2.37.1.windows.1

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app
$ |
```

2. `git status` → Shows the current state of the working directory and staging area, including modified, new, and staged files

3. `git init` → Creates a Git repository

```
91738@IDEAPAD MINGW64 ~/Desktop/hiring-app
$ git status
fatal: not a git repository (or any of the parent directories): .git

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app
$ git init
Initialized empty Git repository in C:/Users/91738/Desktop/hiring-app/.git/

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ ls -a
./ ../ .git/ dockerfile

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$
```

4. `git config --list` → Shows Git username and email

5. `git config --global user.name "<username>"` → Sets Git username

6. git config --global user.email "<email>" → Sets Git email

```
91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git config --global user.name "vamshikumar-a"

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git config --global user.email "vamshikumara1004@gmail.com"

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager-core
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.email=vamshikumara1004@gmail.com
user.name=vamshikumar-a
user.password=738652Vams
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ |
```

```
91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        a
        dockerfile

nothing added to commit but untracked files present (use "git add" to track)

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ |
```

7. git add <filename> → Stages one file

8. git add . → Stages all files

```

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        a
        dockerfile

nothing added to commit but untracked files present (use "git add" to track)

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git add a dockerfile

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   a
        new file:   dockerfile

```

9. git reset → Unstages files

```

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   a
        new file:   dockerfile

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git reset

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        a
        dockerfile

nothing added to commit but untracked files present (use "git add" to track)

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)

```

```

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   a
        new file:   dockerfile

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        b
        c
        d

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git add .

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   a
        new file:   b
        new file:   c
        new file:   d
        new file:   dockerfile

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)

```

10. `git commit -m "message"` → Saves changes as a commit

```

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   a
        new file:   b
        new file:   c
        new file:   d
        new file:   dockerfile

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git commit -m "added files"
[master (root-commit) ed01cab] added files
5 files changed, 2 insertions(+)
create mode 100644 a
create mode 100644 b
create mode 100644 c
create mode 100644 d
create mode 100644 dockerfile

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master
nothing to commit, working tree clean

```

11. git log → Shows commit history

```

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git log
commit ed01cab7ecb863483612c8484b1f4dce2332bb87 (HEAD -> master)
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date:   Tue Aug 18 15:58:30 2026 +0530

    added files

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$

```

12. git push -u origin master → Uploads code to remote repository.


```

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git push -u origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 298 bytes | 149.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:vamshikumar-a/techie.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)

```

13. git remote add origin <remote-repo-url> → Adds remote GitHub repository

```

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git remote add origin git@github.com:vamshikumar-a/techie.git

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git push -u origin master
The authenticity of host 'github.com (20.207.73.82)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdKr4UvCoqU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
git@github.com: Permission denied (publickey).
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/c/Users/91738/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/91738/.ssh/id_rsa
Your public key has been saved in /c/Users/91738/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:kd+sF6QBqRmRyisijgxSvvAj8ys9KMMSKCKgXhCJEZO 91738@IDEAPAD
The key's randomart image is:
+---[RSA 3072]-----+
|++o+..|
|.oE ..|
|...+ +.|
|oo+ B|
|+oo .S|
|#... .o|
|/* ..|
|X=* .|
|o*o+|
+---[SHA256]-----+

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ cat /c/Users/91738/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQgQC5IE0T+Z4wu9FaGiX04Gf7hLKdrM0qpqqXrU8Fes0NxJ9tWZJ7IhhDAH2VzVjQmUdK86

```

14. git remote -v → Shows remote repository URL

```

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git remote -v
origin  git@github.com:vamshikumar-a/techie.git (fetch)
origin  git@github.com:vamshikumar-a/techie.git (push)

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$

```

```

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ ls
a b c d dockerfile

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ rm dockerfile

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ ls
a b c d

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    dockerfile

no changes added to commit (use "git add" and/or "git commit -a")

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ |

```

15. git pull → Gets latest changes from remote

```

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ ls
a abc b c d

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git pull
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 920 bytes | 41.00 KiB/s, done.
From github.com:vamshikumar-a/techie
   3f7289d..9c4be60  master    -> origin/master
Merge made by the 'ort' strategy.
 new-file | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 new-file

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git push -u origin master
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 457 bytes | 457.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To github.com:vamshikumar-a/techie.git
   9c4be60..e32803e  master -> master
branch 'master' set up to track 'origin/master'.

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ ls
a abc b c d new-file

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ |

```

16. git clone <remote-repo-url> → Downloads a remote repository from github

```

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git clone git@github.com:vamshikumar-a/techie.git
Cloning into 'techie'...
remote: Enumerating objects: 13, done.
remote: Counting objects: 100% (13/13), done.
remote: Compressing objects: 100% (7/7), done.
remote: Total 13 (delta 5), reused 9 (delta 3), pack-reused 0 (from 0)
Receiving objects: 100% (13/13), done.
Resolving deltas: 100% (5/5), done.

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ ls
a abc b c d new-file techie/

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$

```

```

91738@IDEAPAD MINGW64 ~/Desktop
$ git clone git@github.com:vamshikumar-a/techie.git
Cloning into 'techie'...
remote: Enumerating objects: 13, done.
remote: Counting objects: 100% (13/13), done.
remote: Compressing objects: 100% (7/7), done.
remote: Total 13 (delta 5), reused 9 (delta 3), pack-reused 0 (from 0)
Receiving objects: 100% (13/13), done.
Resolving deltas: 100% (5/5), done.

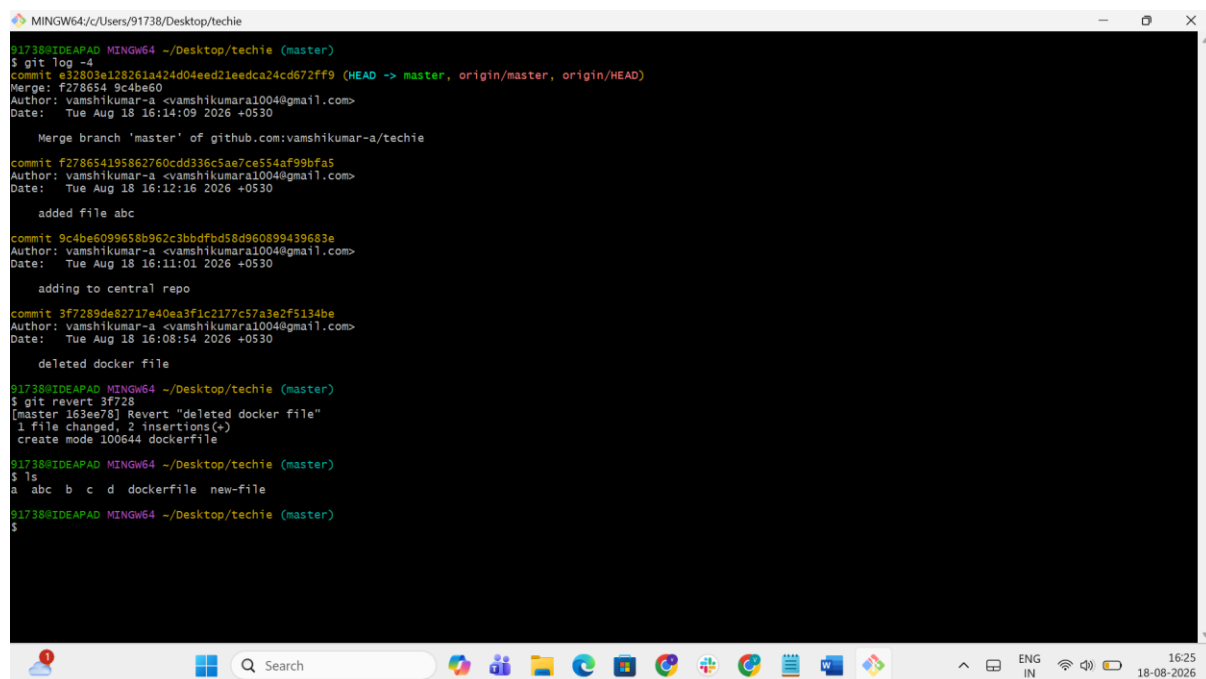
91738@IDEAPAD MINGW64 ~/Desktop
$ cd techie

91738@IDEAPAD MINGW64 ~/Desktop/techie (master)
$ ls
a abc b c d new-file

91738@IDEAPAD MINGW64 ~/Desktop/techie (master)
$

```

17. `git revert <commit_id>` → Undoes a commit safely.



```

MINGW64/C:/Users/91738/Desktop/techie
91738@IDEAPAD MINGW64 ~/Desktop/techie (master)
$ git log -4
commit e32803e128261a24d04eed21eedca24cd672ff9 (HEAD -> master, origin/master, origin/HEAD)
Merge: f278654 9c4be60
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Tue Aug 18 16:14:09 2026 +0530

    Merge branch 'master' of github.com:vamshikumar-a/techie

commit f278654195862760cdd336c5ae7ce554af99bfa5
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Tue Aug 18 16:12:16 2026 +0530

    added file abc

commit 9c4be6099658b962c3bbdfbd58d960899439683e
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Tue Aug 18 16:11:01 2026 +0530

    adding to central repo

commit 3f7289de82717e40ea3f1c2177c57a3e2f5134be
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Tue Aug 18 16:08:54 2026 +0530

    deleted docker file

91738@IDEAPAD MINGW64 ~/Desktop/techie (master)
$ git revert 3f728
[master 163ee78] Revert "deleted docker file"
1 file changed, 2 insertions(+)
create mode 100644 dockerfile

91738@IDEAPAD MINGW64 ~/Desktop/techie (master)
$ ls
a abc b c d dockerfile new-file

91738@IDEAPAD MINGW64 ~/Desktop/techie (master)
$

```

```

91738@IDEAPAD MINGW64 ~/Desktop/techie (master)
$ git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   alphabet

91738@IDEAPAD MINGW64 ~/Desktop/techie (master)
$ git reset

91738@IDEAPAD MINGW64 ~/Desktop/techie (master)
$ git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        alphabet

nothing added to commit but untracked files present (use "git add" to track)
91738@IDEAPAD MINGW64 ~/Desktop/techie (master)

```

```

91738@IDEAPAD MINGW64 ~/Desktop/techie (master)
$ git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        alphabet

nothing added to commit but untracked files present (use "git add" to track)

91738@IDEAPAD MINGW64 ~/Desktop/techie (master)
$ git add .

91738@IDEAPAD MINGW64 ~/Desktop/techie (master)
$ git commit -m "adding alphabet file and revert of docker file"
[master bb13773] adding alphabet file and revert of docker file
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 alphabet

91738@IDEAPAD MINGW64 ~/Desktop/techie (master)
$ git push -u origin master
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 576 bytes | 288.00 KiB/s, done.
Total 5 (delta 2), reused 1 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To github.com:vamshikumar-a/techie.git
   e32803e..bb13773  master -> master
branch 'master' set up to track 'origin/master'.

91738@IDEAPAD MINGW64 ~/Desktop/techie (master)

```